

Overheads of Hardware vs. Software Virtualization

A quantitative analysis of KVM and QEMU TCG Memory subsystems

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Semester 4042

Background

- Virtualization is widely-used in cloud computing, OS development, etc.
- Allows you to run multiple *guest operating systems* inside a *host operating system*.

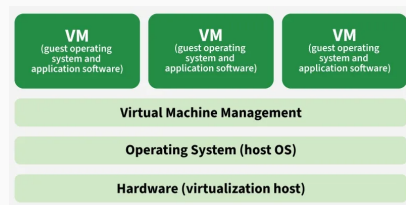


Figure 1: Virtualization

- **Hardware-Assisted Virtualization:**

- Powered by KVM in the Linux kernel for virtualization on the same host architecture.
- Processing is done on actual hardware via CPU Virtualization technologies (VT-x/AMD-V).

- **Emulation:**

- For running OSes that run on entirely different CPU architectures!
- Powered by QEMU's TCG (Tiny Code Generator).

Motivation: The Virtual Memory Challenge

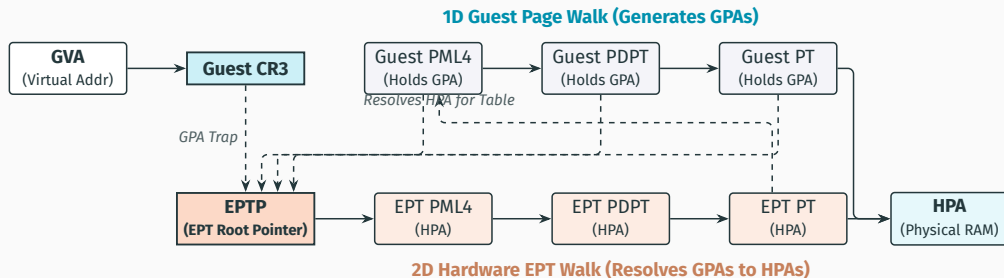
The Guest Memory Layering Illusion:

- Guest OS maps Guest Virtual Addresses (GVA) to Guest Physical Addresses (GPA).
- Host OS must map GPA to final Host Physical Addresses (HPA).

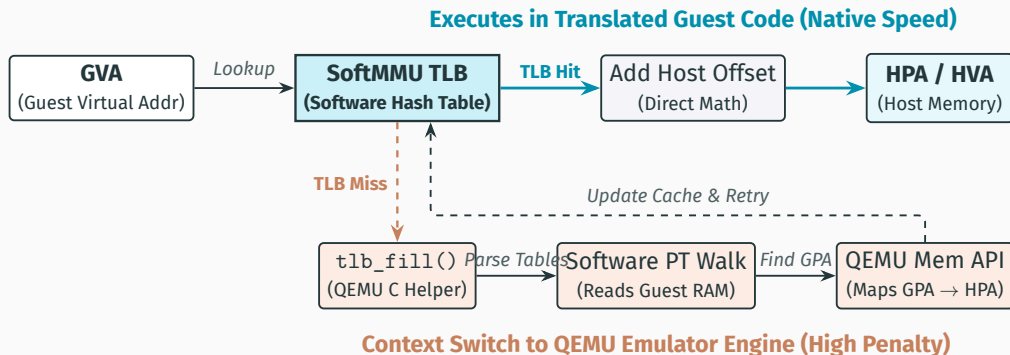
Two Architectural Solutions:

- **Hardware EPT (KVM):** The hardware MMU natively walks both page tables simultaneously in silicon. High-speed, but TLB misses are costly.
- **Software MMU (TCG):** QEMU maintains an inline software TLB cache. If it misses, it triggers a slow, multi-layered C helper routine loop.

EPT Overall Look



SoftMMU Overall Look



Motivation: Instruction Pipeline Divergence

Instruction Handling Disparity:

- **KVM Direct Execution:** Guest instructions execute natively on physical execution units at full hardware frequency.
- Overheads only trigger during trapping instructions (*VM-Exits*) like I/O or hypercalls.

TCG Binary Translation:

- Guest binary blocks are completely intercepted, compiled to Intermediate Representation (IR), and rewritten into Host Assembly.
- Code changes require synchronous, costly generation cache invalidations (*SMC Penalties*).

Project Goal: Quantifying the Virtualization Tax

- **The Main Objective:** Isolate and measure the exact architectural overheads of hardware-assisted memory management (KVM/EPT) versus software emulation (QEMU/SoftMMU).
- **The Telemetry Stack:** Utilize Linux profiling tools to expose cycle-level bottlenecks:
 - `perf stat` for hardware counters (cache misses, dTLB stalls).
 - `perf record` and `perf memo` to map execution hotspots.
 - **Flamegraphs** to visually pinpoint wasted CPU cycles in the virtualization stack.
- **Workload Strategy:** Deploy targeted benchmarks (Pointer Chaser, Matrix Multiplication, Self-Modifying Code) to explicitly stress memory latency, compute boundaries, and JIT translation penalties.

Experimental Setup: The Shielded Environment

- **Host Configuration & CPU Isolation:**

- Dual-core CPU pinning and isolation via kernel command-line parameters
- Hyper-threading and IRQ handling explicitly disabled for those cores to eliminate L1/L2 cache pollution.
- *For further*

- **Guest VM Infrastructure:**

- Orchestrating Debian (NoCloud images) via custom QEMU/KVM scripts.
- **Mixed-Architecture Target:**
 - **x86_64:** Native KVM vs. TCG Emulation.
 - **ARM (AArch64):** Cross-Architecture TCG Emulation.

- **Telemetry Pipeline:**

- Fully automated data collection communicating via SSH.
- `perf` executed on host to preserve isolated guest execution.

Isolation details & scripts: <https://github.com/Tahabik/M-C-Analysis>

Workload Methodology: The Benchmarks

- **Pointer Chaser (Memory Latency Bound):**

- Traverses a linked list with randomized and sequential memory allocations.
- **Stress Profile:** Maximizes dTLB misses and forces constant cache eviction to evaluate the raw penalty of EPT page walks versus SoftMMU table parsing.

- **Matrix Multiplication (Compute & Cache Bound):**

- Performs dense linear algebra operations ($O(n^3)$ complexity).
- **Stress Profile:** Exhibits high spatial and temporal data locality. Tests how well each subsystem handles streaming data across L1, L2, and L3 cache boundaries.

- **Self-Modifying Code (Translation Bound):**

- Generates or alters instructions at runtime before executing them.
- **Stress Profile:** Directly forces the invalidation of JIT-compiled translation blocks. Exposes the execution penalty of QEMU TCG code generation cache flushes.

Benchmark 1 Analysis Roadmap: Pointer Chaser

- **Phase 1: Microarchitectural Profiling** (`perf stat`)
 - Baseline evaluation of Sequential vs. Random access patterns.
 - Hard metrics tracking: IPC, dTLB stalls, and LLC miss ratios.
- **Phase 2: Memory Characterization** (`perf mem & report`)
 - Mapping hit distributions across the cache hierarchy ($L1 \rightarrow L2 \rightarrow L3 \rightarrow \text{DRAM}$).
 - Isolating the assembly-level stall points inside the dereference loop.
- **Phase 3: Macro-Execution Profile (Flamegraphs)**
 - Visualizing KVM kernel violation traps versus TCG engine execution.
 - Quantifying the exact ratio of JIT execution to `tlb_fill()` C helper overrides.

Phase 1: Microarchitectural Profiling (`perf stat`)

Core Metric Insights:

- **IPC Collapse:** Track the exact memory footprint size where IPC drops (e.g., when transitioning from L3 cache to DRAM).
- **The Translation Tax:** Contrast the massive instruction expansion in TCG versus KVM's near 1:1 execution mapping.
- **TLB vs. Cache Violations:** Pinpoint the size boundary where `dTLB-load-misses` spike for both environments.

Architectural Comparative Summary:

- **Sequential Access:** Hardware prefetchers effectively hide memory latency across both x86 and ARM in KVM.
- **Random Access (Stress):** TCG incurs high software TLB miss costs; KVM suffers from costly 2D hardware page walks.
- **Instruction Stream:** TCG shows significantly higher L1-icache and iTLB misses due to dynamic JIT code generation.

Phase 1: Microarchitectural Profiling: KVM vs. TCG (perf stat)

Pointer Chaser Telemetry at Extremes (L1d vs. DRAM Bound)

Metric	x86 KVM (Hardware)	ARM TCG (SoftMMU)	Multiplier / Insight
Baseline Latency (16KB Seq)	1.13 ns	3.81 ns	3.3x (SoftMMU Base Overhead)
Stress Latency (64MB Rand)	61.09 ns	75.92 ns	Gap narrows at physical DRAM limit
Total Instructions Executed	0.92 Billion	29.74 Billion	32x Instruction Bloat in TCG
L1 iCache Misses (64MB Rand)	1.4 Million	140.4 Million	100x Thrashing from JIT/C Helpers
dTLB Misses (64MB Rand)	38,000	482,000	Heavy software page-table parsing

Key Architectural Takeaways:

- **Instruction Expansion:** TCG generates massive instruction bloat to maintain software MMU hash tables compared to KVM's native 2D page walk.
- **Pipeline Pollution:** TCG dynamic translation completely destroys instruction locality, resulting in 100x more instruction cache misses.
- **The Memory Wall:** Both virtualization strategies hit severe bottlenecks at 16MB, indicating host L3 cache overflow.

Phase 2 Setup: Contextual Latency Profiling

The Tool: `perf mem`

- **Beyond Counting:** Shifts from counting raw misses to capturing the exact *latency* (cycles) and *source* (L1, L2, RAM) of specific memory loads.
- **High-Latency Filtration:** We configure the tracer to ignore fast L1 hits and strictly log memory accesses taking > 30 CPU cycles.

The Objective: Hunting the Anomaly

- **Hypothesis Testing:** If hardware EPT is truly faster than SoftMMU, the physical location of the latency will look fundamentally different.
- **The SoftMMU Hook:** We are actively hunting for a highly specific, severe latency signature in TCG that *does not originate from the guest code*.

Phase 2 Results: The Source of Latency (perf mem)

High-Latency Load Distribution (> 30 CPU Cycles)

Hardware EPT (KVM)

Bottleneck Symbol	Subsystem
kvm_lapic_find...	Interrupt Routing
eventfd_poll	Host Kernel I/O
update_curr	Host Scheduler
gsi_handler	QEMU Event Loop

Insight: Hardware page walks are fast enough that the primary latency shifts to virtual interrupt handling and host context switches.

Software MMU (TCG)

Bottleneck Symbol	Subsystem
tlb_fill	SoftMMU Fallback
cpu_mmu_lookup	SoftMMU Hash Table
helper_le_ldq_mmu	JIT Memory Helper
address_space_...	QEMU Mem API

The Anomaly: The SoftMMU data structures themselves are evicted to physical RAM. Emulation stalls parsing its own page tables.

Phase 3 Setup: Visualizing the Execution Stack

The Tool: Call Graphs (`perf record -g`)

- **High-Frequency Sampling:** Captures the host CPU's exact execution stack the moment a hardware cache miss is triggered.
- **Chain of Custody:** Reconstructs the exact chain of function calls (who called whom) leading directly to the pipeline stall.

The Objective: The Flamegraph Abstraction

- **Macro-Level Translation:** Converting millions of dense, textual stack traces into visual, proportional execution blocks.
- **Isolating the Virtualization Tax:** Visually separating cycles spent executing guest memory accesses from cycles burned maintaining the emulator's infrastructure.

Next: Mapping the real-world traces against these expectations.

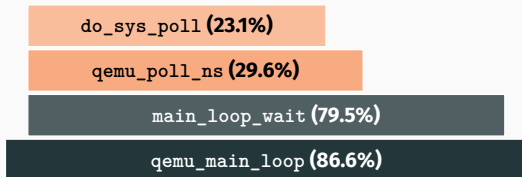
Phase 3: KVM Profile — The Polling Dominance

Host Stack Analysis:

- **The Event Loop Trap:** 86.6% of cache-miss samples originate from the core `qemu_main_loop`.
- **Syscall Bottleneck:** Drills straight down into `main_loop_wait` (79.5%) and `ppoll` (29.6%).
- **Kernel Tax:** 23.1% of stalls occur deep within the host kernel's `do_sys_poll`.

Takeaway: Guest code execution is so fast that hardware stalls shift completely to host I/O multiplexing and kernel context switches.

Dominant Execution Path:



Phase 3: TCG Profile — The JIT Cache Pressure

Host Stack Analysis:

- **JIT Domination:** Nearly **80% (79.96%)** of all pipeline stalls happen directly inside the runtime JIT code buffer.
- **Translation Chaining:** Significant cycles are spent in `helper_lookup_tb_ptr` attempting to chain execution blocks together without exiting to the main loop.
- **Cache Thrashing:** High instruction-cache misses occur because the host CPU is constantly swinging between executing guest code and running SoftMMU lookup helpers.

Dominant Execution Path:

`helper_access_check...` (5%) `helper_lookup_tb_ptr` (3.8%)

JIT Translated Code Cache (79.96%)

Takeaway: Unlike KVM, TCG spends almost zero time polling. The host CPU is entirely bottle-necked compiling, executing, and resolving addresses for guest instructions.

Benchmark 2 results and analysis

Benchmark 3 results and analysis

Benchmark 4 results and analysis